Object Oriented Programming

Project Report

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Project Report: Virtual Escape Room Game (C++ OOP)

1. Project Title

Virtual Escape Room Game using Object-Oriented Programming (OOP) in C++ in GUI

2. Objective

The goal of this project is to develop an interactive, text-based Escape Room game using core Object-Oriented Programming concepts in C++. The game challenges players to solve a variety of puzzles — including logic riddles, ciphers, and word scrambles — within limited attempts to simulate an escape room experience.

3. Key Features

- **Mode Selection**: Players can choose between two major gameplay modes:
  - **Puzzle-based mode** (Crack The Code, Cipher Code, Word Unscramble, Logic Puzzle)
  - **Statement-based puzzle mode**
- **Difficulty Levels**: Easy, Medium, and Hard difficulty levels are provided for puzzle-based mode.
- **Hint System**: Players can type `"hint"` to get a clue for any puzzle.
- **Trap Mechanism**: 3 wrong attempts trigger a failure condition.
- **Player Feedback**: At the end of the game, the user is prompted for feedback and their experience.
- **ASCII Art**: Engaging visuals using ASCII art for welcome and ending messages.

4. OOP Concepts Applied

OOP Concept	Implementation Example
Abstraction	Abstract class `Puzzle` with pure virtual function `display()`
Inheritance	Derived classes: `CrackTheCode`, `CipherCode`, `WordUnscramble`, `LogicPuzzle` from base class `Puzzle`
Polymorphism	Function overriding of `display()` in derived classes
Encapsulation	Puzzle attributes (`question`, `answer`, `hint`) kept protected
Dynamic Binding	Use of virtual functions to resolve method calls at runtime

5. Class Design

a) Puzzle (Abstract Base Class)

- **Attributes**: `question`, `answer`, `hint`
- **Methods**: `display()` (pure virtual), `checkAnswer()`, `getHint()`

b) Derived Puzzle Classes

- `CrackTheCode`, `CipherCode`, `WordUnscramble`, `LogicPuzzle`
- Each overrides the `display()` method.

c) DifficultyLevel (Base Class for Game Modes)

- Contains an array of `Puzzle*`, tracks wrong attempts, handles game loop.
- `play()` method manages puzzle flow, input, hint system, and result.
- Virtual function `loadPuzzles()` to be defined in child classes.

d) Mode Classes

- `EasyMode`, `MediumMode`, `HardMode`, `StatementPuzzleMode`
- Each implements `loadPuzzles()` to populate the puzzle array with appropriate challenges.

6. User Flow

1. Welcome message is displayed.
2. User selects game mode (Puzzle-based or Statement-based).
3. For puzzle-based mode, a difficulty level is selected.
4. The selected game mode loads puzzles via `loadPuzzles()`.
5. User plays through 7 puzzles:
   - Gets hints if needed.
   - Three wrong attempts end the game early.
6. Upon completion, user gives feedback and sees a thank-you message.

7. Technologies Used

- **Programming Language**: C++
- **Compiler**: Any standard C++ compiler (e.g., GCC, MSVC)
- **Development Environment**: Visual Studio Code / Code::Blocks / Qt Creator (for future GUI enhancement)

10. Conclusion

This project successfully demonstrates the power and flexibility of Object-Oriented Programming in C++. The Escape Room game is not only an engaging way to challenge logical thinking but also a practical application of class hierarchies, abstraction, inheritance, and polymorphism. It lays a solid foundation for further development into a full-fledged desktop or mobile puzzle game.

CODE:

```cpp
#include <iostream>
#include <string>

using namespace std;

// Abstract base class (Abstraction)
class Puzzle {
protected:
    string question;
    string answer;
    string hint;

public:
    Puzzle(string q, string a, string h) : question(q), answer(a), hint(h) {}
    virtual void display() = 0; // Pure virtual function
    virtual bool checkAnswer(string userAnswer) {
        return userAnswer == answer;
    }
    virtual string getHint() {
        return hint;
    }
    virtual ~Puzzle() {}
};

// Derived classes (Inheritance + Polymorphism)
class CrackTheCode : public Puzzle {
public:
    CrackTheCode(string q, string a, string h) : Puzzle(q, a, h) {}
    void display() override {
        cout << "Crack the Code: " << question << endl;
    }
};

class CipherCode : public Puzzle {
public:
    CipherCode(string q, string a, string h) : Puzzle(q, a, h) {}
    void display() override {
        cout << "Cipher Puzzle: " << question << endl;
    }
};

class WordUnscramble : public Puzzle {
public:
    WordUnscramble(string q, string a, string h) : Puzzle(q, a, h) {}
    void display() override {
        cout << "Unscramble the Word: " << question << endl;
    }
};

class LogicPuzzle : public Puzzle {
public:
    LogicPuzzle(string q, string a, string h) : Puzzle(q, a, h) {}
    void display() override {
        cout << "Logic Puzzle: " << question << endl;
```

```
    }
};

// Difficulty base class
class DifficultyLevel {
protected:
    Puzzle* puzzles[7];
    int wrongAttempts;

public:
    DifficultyLevel() : wrongAttempts(0) {
        for (int i = 0; i < 7; ++i)
            puzzles[i] = nullptr;
    }
    virtual void loadPuzzles() = 0; // Abstraction
    virtual void play() {
        for (int i = 0; i < 7; i++) {
            puzzles[i]->display();
            string answer;
            cout << "Your Answer: ";
            getline(cin, answer);

            if (answer == "hint") {
                cout << "Hint: " << puzzles[i]->getHint() << endl;
                i--;
                continue;
            }

            if (puzzles[i]->checkAnswer(answer)) {
                cout << "Correct!\n\n";
            }
            else {
                wrongAttempts++;
                cout << "Wrong!\n";
                if (wrongAttempts >= 3) {
                    cout << "You triggered a trap and failed the escape room!\n";
                    return;
                }
                cout << "Type 'hint' if you want a clue.\n";
                i--;
            }
        }

        cout << R"(
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   ||     ||||||   ||||||      |||   ||| || ||   |||| )";

        string feedback, experience;
        cout << "\n\nWe value your feedback!\n";
        cout << "How was your experience? ";
        getline(cin, experience);
        cout << "Any feedback or suggestions? ";
        getline(cin, feedback);
        cout << "\nThanks for your input!\n";
    }
```

```
    virtual ~DifficultyLevel() {
        for (int i = 0; i < 7; ++i)
            delete puzzles[i];
    }
};

// Easy Mode
class EasyMode : public DifficultyLevel {
public:
    void loadPuzzles() override {
        puzzles[0] = new CrackTheCode("Find the next number in the sequence: 2, 4, 
8, 16, ?", "32", "It's a power of 2.");
        puzzles[1] = new WordUnscramble("Unscramble: LPEAP", "Apple", "Think of a 
common fruit.");
        puzzles[2] = new CipherCode("Decode this Caesar cipher with shift 1: UIF", 
"The", "Shift each letter one back in the alphabet.");
        puzzles[3] = new LogicPuzzle("What has to be broken before you can use it?", 
"Egg", "It's found in your kitchen.");
        puzzles[4] = new WordUnscramble("Unscramble: ANANAB", "Banana", "Another 
yellow fruit.");
        puzzles[5] = new CrackTheCode("What's next? 1, 3, 6, 10, 15, ?", "21", 
"Think of triangular numbers.");
        puzzles[6] = new LogicPuzzle("What comes once in a minute, twice in a 
moment, but never in a thousand years?", "M", "Focus on the letters in the words.");
    }
};

// Medium Mode
class MediumMode : public DifficultyLevel {
public:
    void loadPuzzles() override {
        puzzles[0] = new CrackTheCode("Complete the pattern: 2, 6, 12, 20, ?", "30", 
"Look at the difference pattern.");
        puzzles[1] = new CipherCode("ROT13: CYNPR", "Place", "ROT13 means shift 13 
characters.");
        puzzles[2] = new WordUnscramble("Unscramble: OCREPTUM", "Computer", "A 
device you're using right now.");
        puzzles[3] = new LogicPuzzle("I speak without a mouth and hear without ears. 
I have no body, but I come alive with wind. What am I?", "Echo", "It's a sound.");
        puzzles[4] = new WordUnscramble("Unscramble: KROTWREN", "Network", "Trick 
puzzle. No real word-skip.");
        puzzles[5] = new CrackTheCode("What comes next? 0, 1, 1, 2, 3, 5, ?", "8", 
"Fibonacci sequence.");
        puzzles[6] = new LogicPuzzle("I'm tall when I'm young, and I'm short when 
I'm old. What am I?", "Candle", "Used during power outages.");
    }
};

// Hard Mode
class HardMode : public DifficultyLevel {
public:
    void loadPuzzles() override {
        puzzles[0] = new CrackTheCode("Next in pattern: 1, 4, 9, 16, 25, ?", "36", 
"Think squares.");
        puzzles[1] = new CipherCode("Binary to ASCII: 01001000 01001001", "Hi", 
"Each 8-bit represents a letter.");
```

```cpp
        puzzles[2] = new WordUnscramble("Unscramble: NIAGTMLROHPY", "Polymorphing",
"A coding concept.");
        puzzles[3] = new LogicPuzzle("The more you take, the more you leave behind.
What are they?", "Footsteps", "Think of walking.");
        puzzles[4] = new CrackTheCode("If A=1, B=2... What is the sum of 'ESCAPE'?",
"47", "Add letter values.");
        puzzles[5] = new CipherCode("Atbash cipher for: ZOO", "All", "A becomes Z, B
becomes Y...");
        puzzles[6] = new LogicPuzzle("What disappears the moment you say its name?",
"Silence", "Think carefully before speaking.");
    }
};

// Statement Puzzle Mode
class StatementPuzzleMode : public DifficultyLevel {
public:
    void loadPuzzles() override {
        puzzles[0] = new LogicPuzzle("I have cities but no houses, forests but no
trees, and rivers but no water. What am I?", "Map", "Think of something you use to
navigate.");
        puzzles[1] = new CrackTheCode("A broken clock shows 9:13. What 4-digit code
does it represent?", "0913", "The time is correct twice a day.");
        puzzles[2] = new LogicPuzzle("ɾʰʇ sʰʇʐ ɔʇ ʰʇʰʰʇ", "The thorn is walking",
"Use a mirror or reverse the letters.");
        puzzles[3] = new WordUnscramble("ROYGBIV", "ROYGBIV", "Colors of the rainbow
in reverse.");
        puzzles[4] = new CrackTheCode("First of sun, second of fire, third of night,
fourth of sky. What code do the initials make?", "SFNS", "Take the first letter of
each word.");
        puzzles[5] = new CrackTheCode("Magic square: Fill the missing number. 8 1 6,
3 5 7, 4 9 ?", "2", "All rows and columns add up to 15.");
        puzzles[6] = new LogicPuzzle("You entered the Echo Chamber. The room's name
is the password. What is it?", "Echo", "Say the room's name aloud.");
    }
};

void showWelcomeMessage() {
    cout << R"(
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'   `'  '-'  '|  |   |  |
 ______' `_____'    `_____'  `__' `__'__'      `______'       `__' '__'     `_____'
`_____' `__'   `__'
)";
```
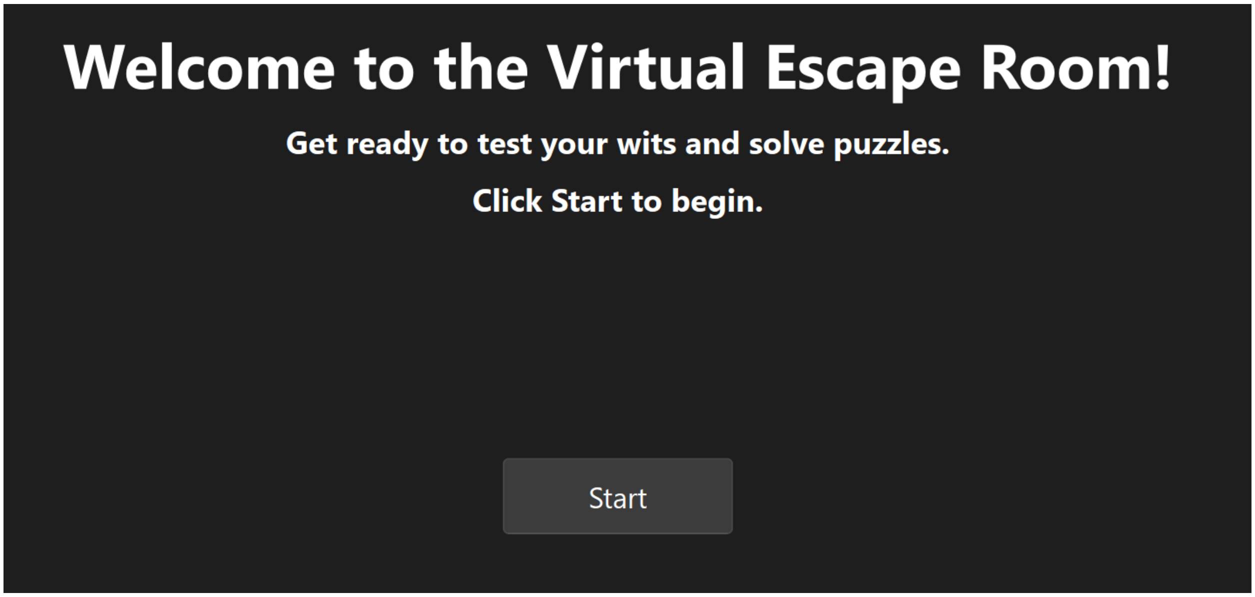
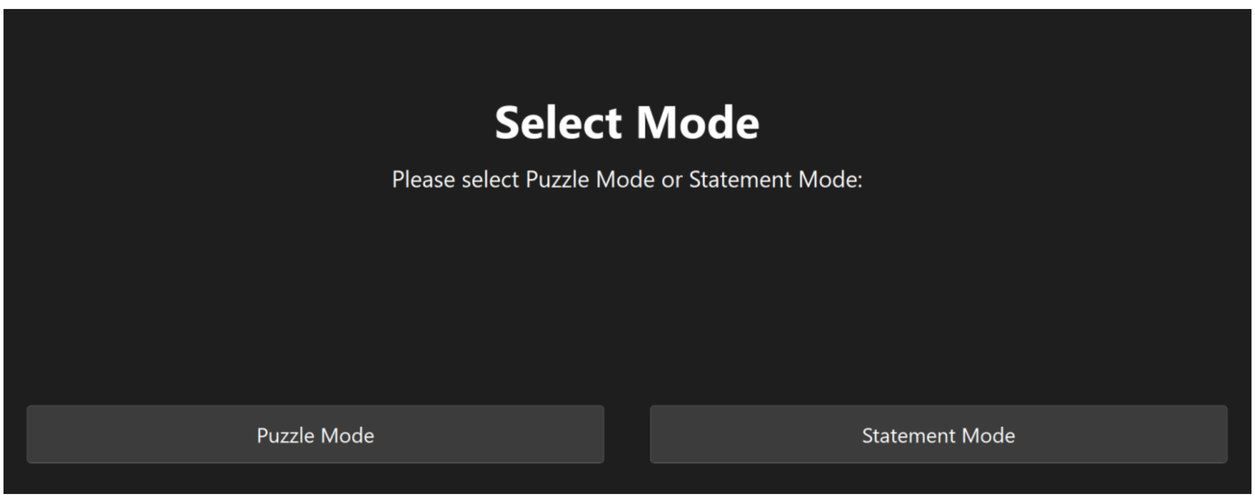
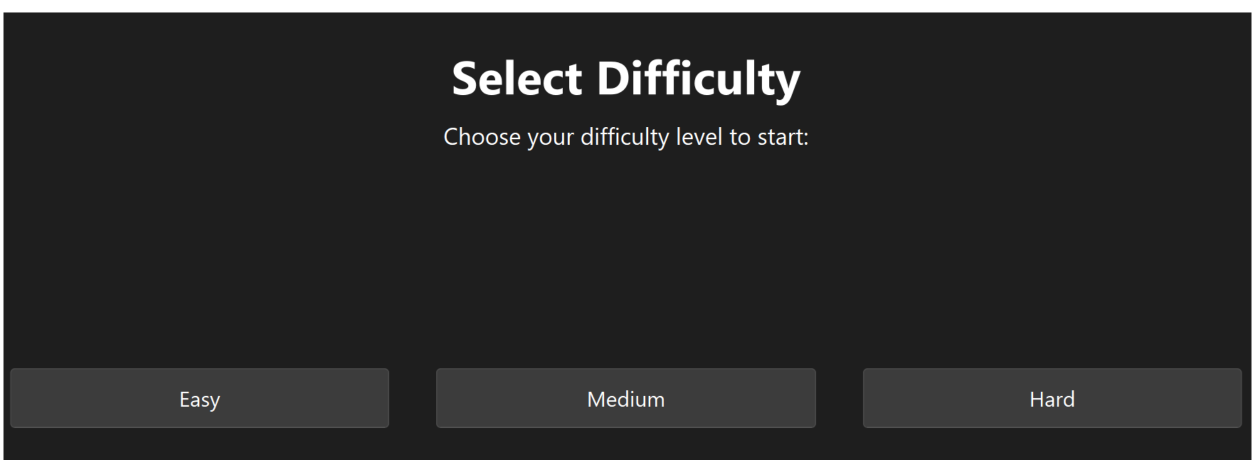
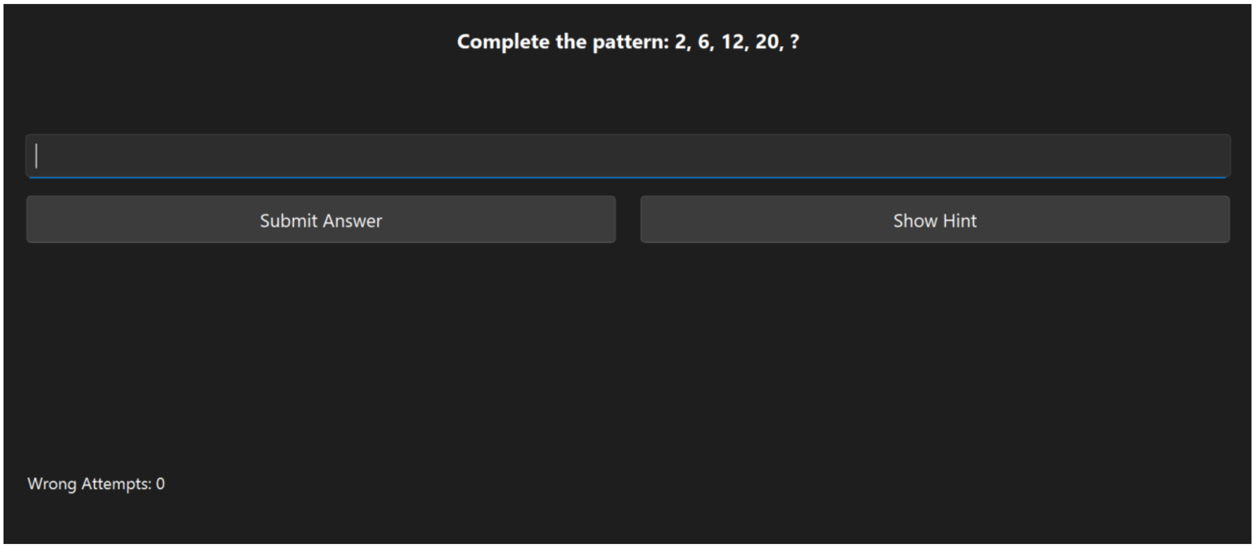

```
    cout << "===== Welcome to the Virtual Escape Room Challenge =====\n";
}

void showEndingMessage() {
    cout << R"END(
**************************************************
*       Thank you for playing ESCAPE ROOM!       *
*        May your mind stay ever sharp!          *
**************************************************
)END" << endl;
}

int main() {
    showWelcomeMessage();

    cout << "Choose Mode:\n1 = Puzzle-based (Crack, Cipher, Unscramble, Logic)\n2 = Statement-based Puzzles\nChoice: ";
    int mode;
    cin >> mode;
    cin.ignore();

    DifficultyLevel* level = nullptr;

    if (mode == 1) {
        cout << "Choose Difficulty (1 = Easy, 2 = Medium, 3 = Hard): ";
        int choice;
        cin >> choice;
        cin.ignore();

        switch (choice) {
        case 1:
            level = new EasyMode();
            break;
        case 2:
            level = new MediumMode();
            break;
        case 3:
            level = new HardMode();
            break;
        default:
            cout << "Invalid choice. Exiting.\n";
            return 0;
        }
    }
    else if (mode == 2) {
        level = new StatementPuzzleMode();
    }
    else {
        cout << "Invalid mode selected.\n";
        return 0;
    }

    level->loadPuzzles();
    level->play();

    delete level;
    showEndingMessage();
```

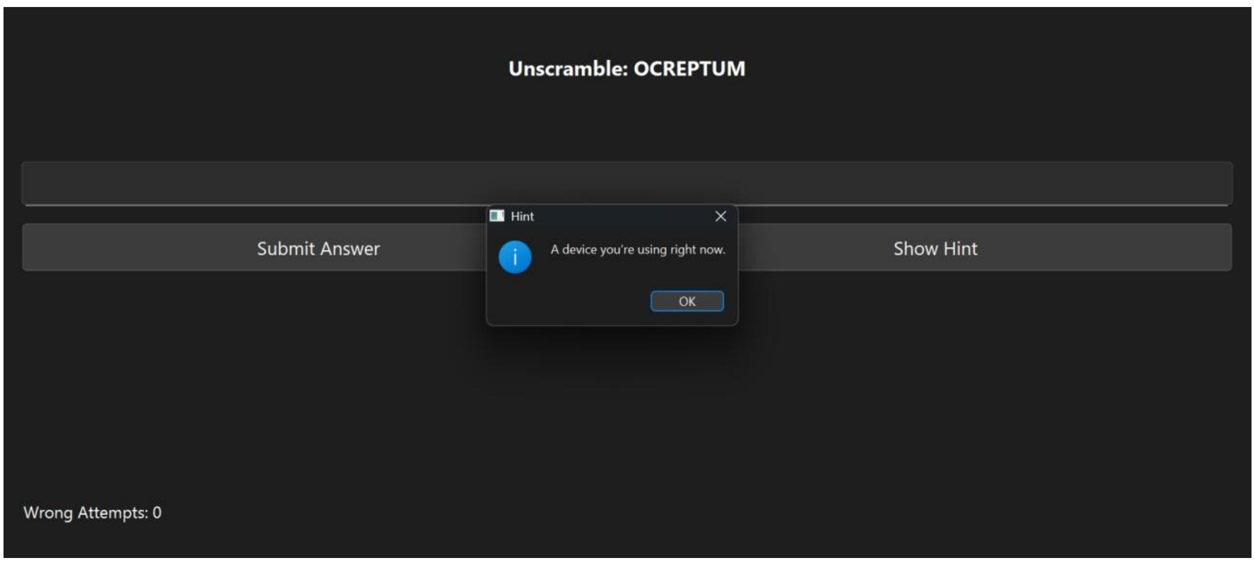
```
    return 0;
}
```

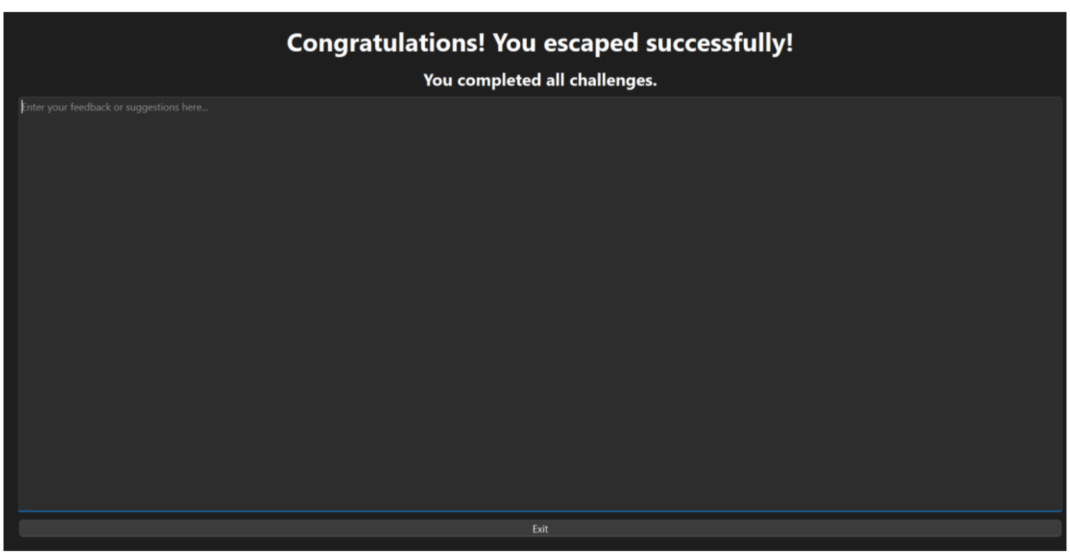

Welcome to the Virtual Escape Room!
Get ready to test your wits and solve puzzles.
Click Start to begin.
Start


Select Mode
Please select Puzzle Mode or Statement Mode:
Puzzle Mode
Statement Mode


Select Difficulty
Choose your difficulty level to start:
Easy
Medium
Hard


Complete the pattern: 2, 6, 12, 20, ?
Submit Answer
Show Hint
Wrong Attempts: 0


Unscramble: OCREPTUM
Submit Answer
Hint
A device you're using right now.
OK
Show Hint
Wrong Attempts: 0


Congratulations! You escaped successfully!
You completed all challenges.
Enter your feedback or suggestions here...
Exit

```
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 `------' `-----'    `-----'  `--' `--'`--'      `------'      `--' '--'     `-----'      `-----' `--'   `--'

===== Welcome to the Virtual Escape Room Challenge =====
Choose Mode:
1 = Puzzle-based (Crack, Cipher, Unscramble, Logic)
2 = Statement-based Puzzles
Choice: 1
Choose Difficulty (1 = Easy, 2 = Medium, 3 = Hard): 2
Crack the Code: Complete the pattern: 2, 6, 12, 20, ?
Your Answer: 30
Correct!

Cipher Puzzle: ROT13: CYNPR
Your Answer: Place
Correct!

Unscramble the Word: Unscramble: OCREPTUM
Your Answer: Computer
Correct!

Logic Puzzle: I speak without a mouth and hear without ears. I have no body, but I come alive with wind. What am I?
Your Answer: Echo
Correct!
```

```
Logic Puzzle: I speak without a mouth and hear without ears. I have no body, but I come alive with wind. What am I?
Your Answer: Echo
Correct!

Unscramble the Word: Unscramble: KROTWREN
Your Answer: NEtword
Wrong!
Type 'hint' if you want a clue.
Unscramble the Word: Unscramble: KROTWREN
Your Answer: Network
Correct!

Crack the Code: What comes next? 0, 1, 1, 2, 3, 5, ?
Your Answer: 8
Correct!

Logic Puzzle: I'm tall when I'm young, and I'm short when I'm old. What am I?
Your Answer: Candle
Correct!


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We value your feedback!
How was your experience?
```